

Juan María Herrera López

Robotics Engineer & Researcher | Machine Learning, Robot Learning & Human-Robot Interaction

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Professional Summary

Robotics and Mechatronics Engineer and PhD researcher developing intelligent robotic systems across machine learning, robot learning, control, and human-robot interaction. Hands-on experience in surgical robotics, perception, learning from demonstration, teleoperation, neuro-symbolic reasoning, and medical-device validation.

Experience

Visiting Robotics Researcher, Bristol Robotics Laboratory (UWE Bristol) – Bristol, UK Sept 2026 – present

- Exploring multimodal AI and vision-language models for robotic perception, action recognition, and task-state understanding.
- Integrating learned perception with structured reasoning and temporal task representations; contributing to experimental design and evaluation.

Robotics Research Engineer / PhD Researcher, IBIMA Plataforma BIONAND / University of Málaga – Málaga, Spain Nov 2022 – present

- Developing machine-learning and neuro-symbolic methods for surgical activity recognition, robotic perception, and high-level reasoning from kinematic and multimodal data.
- Building learning-from-demonstration, trajectory-generation, teleoperation, and force/motion-control pipelines for constrained robotic manipulation using Python, PyTorch, ROS/ROS 2, MATLAB, and Docker.
- Designing reproducible experimental pipelines with user-independent evaluation and contributing to medical-device regulatory documentation, validation, and clinical study protocols.

Engineering Intern - Medical Robotics, Medical Robotics Laboratory, University of Málaga – Málaga, Spain Nov 2021 – June 2022

- Developed a ROS/MATLAB cyber-physical laparoscopic robotics framework and analytical inverse kinematics for a 7-DoF Barrett WAM; work later published as a first-author RIAI article.

Education

University of Málaga, PhD in Mechatronics Engineering – Málaga, Spain Oct 2023 – present

- Human-Robot Collaboration in Minimally Invasive Surgery; FPU Predoctoral Fellowship

University of Málaga, MEng in Mechatronics Engineering – Málaga, Spain Oct 2022 – July 2023

University of Málaga, BEng in Electronics, Robotics and Mechatronics Engineering – Málaga, Spain Oct 2018 – July 2022

- Average grade: 8.81/10; Best Academic Record Award (2021/2022 graduating cohort)

Selected Publications

IEEE Access (2024, first author): [Laparoscopic Suture Gestures Recognition via Machine Learning Robotics and Autonomous Systems \(2026\): Ontology-based action planner for robotic laparoscopic suturing](#)

Applied Sciences (2026): [Decoupling Tissue and Abdominal Forces in Laparoscopic Robotic Surgery via Viscoelastic Modeling](#)

RIAI (2023, first author): [Entorno de trabajo ciber-físico para cirugía laparoscópica](#)

Technical Skills

Robotics: ROS/ROS 2, robot control, teleoperation, HRI, motion planning, force/motion control, learning from demonstration, movement primitives, RCM-constrained manipulation

AI & Machine Learning: Python, PyTorch, deep learning, computer vision, sequence modeling, neuro-symbolic AI, vision-language models, scikit-learn

Software & Engineering: Linux, Docker, Git, MATLAB, C++, reproducible experimental pipelines

Domain: surgical and medical robotics, minimally invasive surgery, kinematic modeling, medical-device validation and regulatory documentation

Languages

Languages: Spanish (native), English (C1 - Trinity College London)